Stanford **Spring 2026**

AA 203: Optimal and Learning-based Control
Homework #1
Due Friday, April 24 by 5:00 pm

Learning goals for this problem set:

Problem 1: To gain insights into the implementation of gradient methods and review some notions of linear algebra.

Problem 2: To familiarize with Linear Quadratic control and learn a first algorithmic approach to this problem.

Problem 3: Become familiar with the process of solving calculus of variations problems.

Problem 4: To familiarize with the Hamiltonian equations for optimal control.

1.1 Gradient descent and line search. Let $Q \in \mathbb{R}^{n\times n}$ be a symmetric positive-definite matrix, and $b \in \mathbb{R}^n$ be a given vector. Consider the quadratic optimization problem

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}x^\top Qx - b^\top x.$$

Let $f(x) \coloneqq \frac{1}{2}x^\top Qx - b^\top x$, and denote the eigenvalues of Q as $\lambda_1, \ldots, \lambda_n$.

(a) Find the unique local minimum candidate $x^* \in \mathbb{R}^n$. Prove x^* is a global minimum.

Hint: Any twice-differentiable function f is strictly convex if the Hessian $\nabla^2 f(x)$ is positive-definite for all $x \in \mathbb{R}^n$.

(b) Show that, starting from any initial point $x^{(0)} \in \mathbb{R}^n$, Newton's method with constant step size $\eta = 1$ converges in one iteration to the optimal solution x^*. Hence, performing one step of Newton's method is equivalent to solving the linear system of equations $Qx = b$. What would be the downside of this solution method if n is large (e.g., $n \gg 10^4$) and the matrix Q has no particular structure?

(c) Let $S \in \mathbb{R}^{n\times n}$ be a symmetric matrix. By the Spectral Theorem, there exist an orthogonal matrix $U \in \mathbb{R}^{n\times n}$ and a diagonal matrix $\Sigma = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ such that $S = U\Sigma U^\top$. Show $\|Sx\|_2 = \|\Sigma U^\top x\|_2$ for any $x \in \mathbb{R}^n$. Then show $\|\Sigma z\|_2 \le \max_{i\in\{1,\ldots,n\}}|\mu_i|\|z\|_2$ for any $z \in \mathbb{R}^n$. Finally, conclude that $\|Sx\|_2 \le \max_{i\in\{1,\ldots,n\}}|\mu_i|\|x\|_2$ for any $x \in \mathbb{R}^n$.

Hint: If $U \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, then $\|Uy\|_2 = \|U^\top y\|_2 = \|y\|_2$ for any $y \in \mathbb{R}^n$.

(d) For any $\eta > 0$, show that the eigenvalues of the matrix $I - \eta Q$ are exactly $\{1 - \eta\lambda_i\}_{i=1}^n$.

Hint: Identify an orthonormal basis of vectors $\{v_i\}_{i=1}^n \subset \mathbb{R}^n$ such that $(I - \eta Q)v_i = (1 - \eta\lambda_i)v_i$ for each i.

(e) Consider the gradient descent update rule $x^{(k+1)} = x^{(k)} - \eta\nabla f(x^{(k)})$ at iteration $k \in \mathbb{N}_{\ge 0}$ with a constant step size $\eta > 0$. Define $\delta^{(k)} \coloneqq \|x^{(k)} - x^*\|_2$ and $\gamma(\eta) \coloneqq \max_{i\in\{1,\ldots,n\}}|1 - \eta\lambda_i|$. Use an inductive argument to show $\delta^{(k)} \le \gamma(\eta)^k\delta_0$ for all $k \in \mathbb{N}_{\ge 0}$.

(a) Find the unique local minimum candidate $x^* \in \mathbb{R}^n$. Prove x^* is a global minimum.

Hint: Any twice-differentiable function f is strictly convex if the Hessian $\nabla^2 f(x)$ is positive-definite for all $x \in \mathbb{R}^n$.

$\nabla f(x) \in \mathbb{R}^n$

$$f(x) = \frac{1}{2} x^T Q x - b^T x$$

$\nabla f(x) = Qx - b$

$\nabla^2 f(x) = Q$ $\quad$ $f(x)$ is strictly convex as Q is PD

Then the local minimum is the global minimum

$\nabla f(x) = Qx - b = \emptyset \quad \therefore \quad x^* = Q^{-1} b$

(b) Show that, starting from any initial point $x^{(0)} \in \mathbb{R}^n$, Newton's method with constant step size $\eta = 1$ converges in one iteration to the optimal solution x^*. Hence, performing one step of Newton's method is equivalent to solving the linear system of equations $Qx = b$. What would be the downside of this solution method if n is large (e.g., $n \gg 10^4$) and the matrix Q has no particular structure?

$$x^{(k+1)} = x^{(k)} - \eta\left(\nabla^2 f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$$

With $\eta = 1$

$$x^{(k+1)} = x^{(k)} - Q^{-1}\left[Qx^{(k)} - b\right] = x^{(k)} - x^{(k)} + Q^{-1}b = Q^{-1}b$$

$x^{(k+1)} = x^*$

Starting from $x^{(0)} \rightarrow x^{(1)} = x^*$

If $n \gg 10^4$ and Q has no structure, then solving the linear system can be expensive

(c) Let $S \in \mathbb{R}^{n\times n}$ be a symmetric matrix. By the Spectral Theorem, there exist an orthogonal matrix $U \in \mathbb{R}^{n\times n}$ and a diagonal matrix $\Sigma = \mathrm{diag}(\mu_1, \ldots, \mu_n)$ such that $S = U\Sigma U^\mathsf{T}$. Show $\|Sx\|_2 = \|\Sigma U^\mathsf{T} x\|_2$ for any $x \in \mathbb{R}^n$. Then show $\|\Sigma z\|_2 \le \max_{i\in\{1,\ldots,n\}} |\mu_i| \|z\|_2$ for any $z \in \mathbb{R}^n$. Finally, conclude that $\|Sx\|_2 \le \max_{i\in\{1,\ldots,n\}} |\mu_i| \|x\|_2$ for any $x \in \mathbb{R}^n$.

Hint: If $U \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, then $\|Uy\|_2 = \|U^\mathsf{T} y\|_2 = \|y\|_2$ for any $y \in \mathbb{R}^n$.

1 $\|Sx\|_2 = \|U\Sigma U^T x\|_2 = \|\Sigma U^T x\|_2$ (as U is orthogonal)

2 $\|\Sigma z\|_2 = \|\Sigma\|_2 \|z\|_2 = \sqrt{\sum_{i,j} \Sigma_{ij}^2} \sqrt{\sum_i z_i^2} = \sqrt{\sum_i \mu_i^2} \cdot \sqrt{\sum_i z_i^2} = \sqrt{\sum_i \mu_i^2 z_i^2}$

Considering $\mu_i^2 \le \max_i |\mu_i|^2 \quad \forall i \in \{1, \ldots, n\}$, then

$$\|\Sigma z\|_2 = \sqrt{\sum_i \mu_i^2 z_i^2} \le \sqrt{\sum_i \left(\max_i |\mu_i|\right)^2 z_i^2} = \max_i |\mu_i| \sqrt{\sum_i z_i^2} = \max_i |\mu_i| \|z\|_2$$

③ $\|Sx\|_2 = \|\Sigma U^T x\|_2 = \|\Sigma x\|_2$.

Then $\|\Sigma x\|_2 \le \max_i |\mu_i| \|x\|_2$

Finally $\|Sx\|_2 \le \max_i |\mu_i| \|x\|_2$

(d) For any $\eta > 0$, show that the eigenvalues of the matrix $I - \eta Q$ are exactly $\{1 - \eta\lambda_i\}_{i=1}^n$.

Hint: Identify an orthonormal basis of vectors $\{v_i\}_{i=1}^n \subset \mathbb{R}^n$ such that $(I - \eta Q)v_i = (1 - \eta\lambda_i)v_i$ for each i.

Since Q is symmetric, it has an orthonormal basis of eigenvectors v_i, $i \in \{i, \cdots, n\}$

Then for each eigenvector v_i, $Qv_i = \lambda_i v_i$

So $(I - \eta Q)v_i = v_i - \eta Q v_i = v_i - \eta \lambda_i v_i = (1 - \eta \lambda_i) v_i \quad \forall\, i \in \{1, \cdots, n\}$

(e) Consider the gradient descent update rule $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$ at iteration $k \in \mathbb{N}_{\geq 0}$ with a constant step size $\eta > 0$. Define $\delta^{(k)} := \|x^{(k)} - x^*\|_2$ and $\gamma(\eta) := \max_{i \in \{1,\ldots,n\}} |1 - \eta\lambda_i|$. Use an inductive argument to show $\delta^{(k)} \leq \gamma(\eta)^k \delta_0$ for all $k \in \mathbb{N}_{\geq 0}$.

Consider $x^* = Q^{-1}b$, so $b = Qx^*$

Then the gradient is

$$\nabla F(x^{(k)}) = Qx^{(k)} - b = Qx^{(k)} - Qx^* = Q(x^{(k)} - x^*)$$

Therefore $x^{(k+1)} = x^{(k)} - \eta Q(x^{(k)} - x^*)$

Then $x^{(k+1)} - x^* = x^{(k)} - x^* - \eta Q(x^{(k)} - x^*) = (I - \eta Q)(x^{(k)} - x^*)$

From ⓓ, eigenvalues of $(I - \eta Q)$ are $(1 - \eta \lambda_i)$ $\forall\, i \in \{1, \cdots, n\}$, so

RHS $\Rightarrow \|I - \eta Q\|_2 \|x^{(k)} - x^*\| = \|I - \eta Q\|_2 \delta^{(k)} = \|I - \eta \lambda\|_2 \delta^{(k)}$

Then, leveraging ⓒ $S = U \Sigma U^T$

where $S = I - \eta Q$, a symmetric matrix,

$U = [v_i \cdots v_n]$ is an orthogonal matrix of eigenvectors,

$\Sigma = \mathrm{diag}(1 - \eta\lambda_i, \cdots, 1 - \eta\lambda_n)$ is a diag. matrix of real eigenvalues

then $\|S(x^{(k)} - x^*)\|_2 = \|I - \eta Q\|_2 \delta^{(k)}$, so

$\|I - \eta Q\|_2 \delta^{(k)} \le \max_i |1 - \eta \lambda_i| \delta^{(k)} = \gamma(\eta) \delta^{(k)}$

Finally $\|x^{(k+1)} - x^*\|_2 = \delta^{(k+1)} = \|I - \eta Q\|_2 \, \delta^{(k)} \leq \gamma(\eta)\, \delta^{(k)}$

$$\underline{\delta^{(k+1)} \leq \gamma(\eta)\, \delta^{(k)}}$$

ITERS

For $k = 0$ $\quad \delta^{(1)} \leq \gamma(\eta)\, \delta^{(0)}$

For $k = 1$ $\quad \delta^{(2)} \leq \gamma(\eta)\, \delta^{(1)} \leq \gamma(\eta)(\gamma(\eta)\, \delta^{(0)}) = \gamma(\eta)^2 \delta^{(0)}$

For $k = 3$ $\quad \delta^{(3)} \leq \gamma(\eta)[\gamma(\eta)(\gamma(\eta)\, \delta^{(0)})] = \gamma(\eta)^3 \delta^{(0)}$

$\vdots$

$k = N$ $\quad \delta^{(N+1)} \leq \gamma(\eta)^{N+1} \delta(0)$

So $\quad \delta^{(k+1)} \leq \gamma(\eta)^{k+1} \delta(0) \quad \forall\, k \in \mathbb{N}_{\geq 0}$

(f) Consider gradient descent with exact line search. At each iteration k, denote the descent direction by $d^{(k)} := -\nabla f(x^{(k)})$ and the optimal step size by

$$\eta^{(k)} := \arg\min_{\eta \geq 0} f(x^{(k)} + \eta d^{(k)}).$$

Prove

$$\eta^{(k)} = \frac{\|d^{(k)}\|_2^2}{{d^{(k)}}^\mathsf{T} Q d^{(k)}}.$$

(g) For $n = 2$ and $f(x) = \frac{1}{2}(x_1^2 + \gamma x_2^2)$ with $\gamma = 10$, what is the optimal solution x^*? Implement gradient descent with a constant step size and exact line search, starting from $x^{(0)} = (5, 1)$ and $x^{(0)} = (1, 5)$. What do you observe with exact line search? When does gradient descent begin to "zig-zag"? What issue do you observe with a constant step size? Repeat both experiments with $\gamma = 1$. Submit your plots.

f) $f(x^{(k)} + \eta d^{(k)}) = \frac{1}{2}(x^{(k)} + \eta d^{(k)})^T Q (x^{(k)} + \eta d^{(k)}) - b^T(x^{(k)} + \eta d^{(k)})$

$= \frac{1}{2}\left[x^{(k)T} Q x^{(k)} + \eta x^{(k)T} Q d^{(k)} + \eta d^{(k)T} Q x^{(k)} + \eta^2 d^{(k)T} Q d^{(k)}\right] - b^T x^{(k)} - \eta b^T d^{(k)}$

Q is symmetric, so

$f(x^{(k)} + \eta d^{(k)}) = \frac{1}{2} x^{(k)T} Q x^{(k)} - b^T x^{(k)} + \eta d^{(k)T} Q x^{(k)} + \frac{1}{2}\eta^2 d^{(k)T} Q d^{(k)} - \eta b^T d^{(k)}$, $b^T d^{(k)} = d^{(k)T} b$

$= f(x^{(k)}) + \eta d^{(k)T}(Q x^{(k)} - b) + \frac{1}{2}\eta^2 d^{(k)T} Q d^{(k)}$

$\nabla_\eta f(x^{(k)} + \eta d^{(k)}) = d^{(k)T}(Q x^{(k)} - b) + \eta d^{(k)T} Q d^{(k)} = d^{(k)T} \nabla_x f(x^{(k)}) + \eta d^{(k)T} Q d^{(k)}$

$= -d^{(k)T} d^{(k)} + \eta d^{(k)T} Q d^{(k)} = -\|d^{(k)}\|_2^2 + \eta d^{(k)T} Q d^{(k)}$

$\nabla_\eta f(x^{(k)} + \eta d^{(k)}) = \emptyset = -\|d^{(k)}\|_2^2 + \eta d^{(k)T} Q d^{(k)}$ $\therefore$ $\eta = \|d^{(k)}\|_2^2 \left(d^{(k)T} Q d^{(k)}\right)^{-1}$

1.2 LQR as a QP. Consider the Linear Time-Invariant (LTI) dynamical system

$$x_{t+1} = Ax_t + Bu_t,$$

where $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$ are given matrices, and $x_t \in \mathbb{R}^n$ and $u_t \in \mathbb{R}^m$ are the system state and applied control input, respectively, at time $t \in \mathbb{N}_{\geq 0}$.

Let $x_0 \in \mathbb{R}^n$ be the fixed initial state and $T \in \mathbb{N}$ be some time horizon. Our goal is to find a sequence of control inputs $u^* := (u_0^*, u_1^*, \ldots, u_{T-1}^*) \in \mathbb{R}^{mT}$ that minimizes the quadratic cost

$$J(u) := x_T^\mathsf{T} Q_T x_T + \sum_{t=0}^{T-1} \Big(x_t^\mathsf{T} Q x_t + u_t^\mathsf{T} R u_t \Big),$$

where $Q_T \in \mathbb{R}^{n\times n}$, $Q \in \mathbb{R}^{n\times n}$, and $R \in \mathbb{R}^{m\times m}$ are positive-definite matrices. Later, we will see how dynamic programming can be used to derive an elegant, recursive solution to this problem. For now, we study a convex least-squares formulation. Specifically, we reformulate the problem of minimizing $J(u)$ as

$$\min_{u\in\mathbb{R}^{mT}} \frac{1}{2} u^\mathsf{T} \tilde{Q} u - \tilde{b}^\mathsf{T} u,$$

where $u := (u_0, u_1, \ldots, u_{T-1}) \in \mathbb{R}^{mT}$ is the vector of stacked control inputs, $\tilde{Q} \in \mathbb{R}^{mT\times mT}$ is a positive-definite matrix, and $\tilde{b} \in \mathbb{R}^{mT}$.

(a) Write down $\tilde{Q}$ and $\tilde{b}$ in terms of Q_T, Q, R, A, B, and x_0.

(b) With this reformulation, implement the gradient descent algorithm of your choice to compute the optimal sequence of control inputs u^* for

$$Q_T = 10I_2, \quad Q = I_2, \quad R = I_1, \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T = 20,$$

where I_n is the identity matrix with dimension n. What is the optimal cost $J(u^*)$?

(a) Write down $\tilde{Q}$ and $\tilde{b}$ in terms of Q_T, Q, R, A, B, and x_0.

$$X_1 = AX_0 + BU_0$$
$$X_2 = A(AX_0 + BU_0) + BU_1 = A^2X_0 + ABU_0 + BU_1$$
$$X_3 = A^3X_0 + A^2BU_0 + ABU_1 + BU_2$$
$$X_4 = A^4X_0 + A^3BU_0 + A^2BU_1 + ABU_2 + BU_3$$

$\hookrightarrow$

$$X = \mathbb{V}x_0 + \mathbb{W}U$$

$$\mathbb{V} = \begin{bmatrix} A & A^2 & A^3 & \cdots & A^T \end{bmatrix}^T, \quad X = \begin{bmatrix} X_1^T, \cdots X_T^T \end{bmatrix}^T, \quad U = \begin{bmatrix} U_0^T, \cdots U_{T-1}^T \end{bmatrix}^T$$

$$\mathbb{W} = \begin{bmatrix} B & 0 & 0 & \cdots & 0 \\ AB & B & 0 & \cdots & 0 \\ A^2B & AB & B & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{T-1}B & A^{T-2}B & A^{T-3}B & \cdots & B \end{bmatrix}$$

$$J = x_0^TQx_0 + x_1^TQ_1x_1 + U_0^TRU_0 \qquad T=1$$
$$= x_0^TQx_0 + x_1^TQx_1 + x_2^TQ_2x_2 + U_0^TRU_0 + U_1^TRU_1 \qquad T=2$$
$$= \underbrace{x_0^TQx_0 + x_1^TQx_1 + x_2^TQ_2x_2 + x_3^TQ_3x_3}_{X^THX} + \underbrace{U_0^TRU_0 + U_1^TRU_1 + U_2^TRU_2}_{U^TNU} \qquad T=3$$

$$H = \mathrm{diag}(Q, Q, \cdots, Q_T), \quad N = \mathrm{diag}(R, \cdots, R)$$

$$J(U) = X^THX + U^TNU$$

$$= (\mathbb{V}x_0 + \mathbb{W}U)^TH(\mathbb{V}x_0 + \mathbb{W}U) + U^TNU$$

$$= (\mathbb{V}x_0)^TH(\mathbb{V}x_0) + 2(\mathbb{V}x_0)^TH(\mathbb{W}U) + (\mathbb{W}U)^TH(\mathbb{W}U) + U^TNU$$

$$= x_0^T\mathbb{V}^TH\mathbb{V}x_0 + 2x_0^T\mathbb{V}^TH\mathbb{W}U + U^T\mathbb{W}^TH\mathbb{W}U + U^TNU$$

$$= U^T(\mathbb{W}^TH\mathbb{W} + N)U + 2x_0^T\mathbb{V}^TH\mathbb{W}U + x_0^T\mathbb{V}^TH\mathbb{V}x_0$$

$$= U^T(\mathbb{W}^TH\mathbb{W} + N)U + 2U^T\mathbb{W}^TH\mathbb{V}x_0 + \underbrace{x_0^T\mathbb{V}^TH\mathbb{V}x_0}_{\text{not dependent on } U.\ \text{Ignored}}$$

So, in the form $\frac{1}{2}U^T\tilde{Q}U - \tilde{b}^TU$:

$$\tilde{Q} = 2(\mathbb{W}^TH\mathbb{W} + N), \quad \tilde{b}^T = -2\mathbb{W}^TH\mathbb{V}x_0$$

1.3 Extremal curves. Given the functional

$$J(x) = \int_0^1 \left(\frac{1}{2}\dot{x}(t)^2 + 5x(t)\dot{x}(t) + x(t)^2 + 5x(t) \right) dt,$$

find an extremal curve $x^* : [0,1] \to \mathbb{R}$ that satisfies $x^*(0) = 1$ and $x^*(1) = 3$.

Euler equation: $g_x(x^*, \dot{x}^*, t) - \frac{d}{dt} g_{\dot{x}}(x^*, \dot{x}^*, t) = \emptyset$

$g(x, \dot{x}, t) = \frac{1}{2}\dot{x}^2 + 5x\dot{x} + x^2 + 5x$

$\frac{\partial}{\partial x} g(x,\dot{x},t) = 5\dot{x} + 2x + 5$

$\frac{\partial}{\partial \dot{x}} g(x,\dot{x},t) = \dot{x} + 5x$

$5\dot{x} + 2x + 5 - \frac{d}{dt}(\dot{x} + 5x) = \emptyset$

$\cancel{5\dot{x}} + 2x + 5 - \ddot{x} - \cancel{5\dot{x}} = \emptyset \quad \therefore \quad -\ddot{x} + 2x + 5 = \emptyset$

$\ddot{x} - 2x = 5$

Non-Homogeneous diff. equation

$x(t) = x_h(t) + x_p(t)$

$r^2 - 2 = \emptyset \quad \therefore \quad r = \pm\sqrt{2}$

$x_h(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t}$

$x_p(t) = C = -\frac{5}{2}$

$\dot{x}_p = \emptyset \;,\; \ddot{x}_p = \emptyset \rightarrow \emptyset - 2C = 5 \quad \therefore \quad C = -\frac{5}{2}$

$x(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t} - \frac{5}{2}$

Using boundary conditions:

$x(\emptyset) = 1 = C_1 e^{\sqrt{2}\cdot\emptyset} + C_2 e^{-\sqrt{2}\cdot\emptyset} - \frac{5}{2} = C_1 + C_2 - \frac{5}{2} \quad \therefore \quad C_1 = \frac{7}{2} - C_2$

$x(1) = 3 = \left(\frac{7}{2} - C_2\right) e^{\sqrt{2}} + C_2 e^{-\sqrt{2}} - \frac{5}{2} = \frac{7}{2} e^{\sqrt{2}} + \left(e^{-\sqrt{2}} - e^{\sqrt{2}}\right) C_2$

$C_2 = \frac{11 - 7e^{\sqrt{2}}}{2(e^{-\sqrt{2}} - e^{\sqrt{2}})} \;,\quad C_1 = \frac{7}{2} - \frac{11 - 7e^{\sqrt{2}}}{2(e^{-\sqrt{2}} - e^{\sqrt{2}})}$

1.4 Zermelo's ship. Zermelo's ship must travel through a region of strong currents. The position of the ship is denoted by $(x(t), y(t)) \in \mathbb{R}^2$. The ship travels at a constant speed $v > 0$, yet its heading $\theta(t)$ can be controlled. The current moves in the positive x-direction with speed $w(y(t))$. The equations of motion for the ship are

$$\begin{aligned} \dot{x}(t) &= v\cos\theta(t) + w(y(t)) \\ \dot{y}(t) &= v\sin\theta(t) \end{aligned} .$$

$\dot{x} = f(\cdot\cdot)$

We want to control the heading $\theta(t)$ such that the ship travels from a given initial position $(x(t_0), y(t_0)) = (x_0, y_0)$ to the origin $(0,0)$ in minimum time.

(a) Suppose $w(y(t)) = \frac{v}{h}y(t)$, where $h > 0$ is a known constant. Show that an optimal control law $\theta^*(t)$ must satisfy a linear tangent law of the form

$$\tan\theta^*(t) = \alpha - \frac{v}{h}t$$

for some constant $\alpha \in \mathbb{R}$.

(b) Suppose $w(y(t)) \equiv \beta$ for some constant $\beta > 0$. Derive an expression for the optimal transfer time $t_1^* - t_0$. Make sure to consider each of the following cases: (i) $v > \beta$, (ii) $v = \beta$, and (iii) $v < \beta$. State restrictions, if any, on x_0 under these conditions.

$J = \int_{t_0}^{t_f} 1 \, dt$

$\dot{x} = v\cos\theta + \frac{v}{h}y$

a) Necessary Condition for Opt:

$H = 1 + p_x\left[v\cos\theta + \frac{v}{h}y\right] + p_y v\sin\theta$

$\dot{x}^* = \frac{\partial H}{\partial p_x} = v\cos\theta + \frac{v}{h}y$

$\dot{y}^* = \frac{\partial H}{\partial p_y} = v\sin\theta$

$\dot{p}_x = -\frac{\partial H}{\partial x} = \phi \therefore p_x = c_1$

$\dot{p}_y = -\frac{\partial H}{\partial y} = -p_x\frac{v}{h} = -c_1\frac{v}{h} \therefore p_y = -c_1\frac{v}{h}t + c_2$

$\varnothing = \frac{\partial H}{\partial \theta} = -p_x v\sin\theta + p_y v\cos\theta \therefore p_x\sin\theta = p_y\cos\theta \therefore \tan\theta^* = \frac{p_y}{p_x} = -\frac{v}{h}t + \frac{c_2}{c_1}$

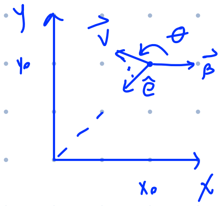
$H = 1 + p_x\left[v\cos\theta + \beta\right] + p_y v\sin\theta$

$\dot{p}_x = -\frac{\partial H}{\partial x} = \phi \therefore p_x = c_1$

$\dot{p}_y = -\frac{\partial H}{\partial y} = \phi \therefore p_y = c_2$

$\varnothing = \frac{\partial H}{\partial \theta} = -p_x v\sin\theta + p_y v\cos\theta \therefore p_x\sin\theta = p_y\cos\theta \therefore \tan\theta^* = \frac{p_y}{p_x} = \frac{c_2}{c_1}$

$\dot{x} = v\cos\theta + \beta$
$\dot{y} = v\sin\theta$



b) Opt. time → max. speed to reach origin

$$t_f^* - t_0 = \frac{r_0}{s}, \quad r_0 = \sqrt{x_0^2 + y_0^2}$$

$$\hat{e} = \left(-\frac{x_0}{r_0}, -\frac{y_0}{r_0}\right)$$

For opt. time, need largest speed in $\hat{e}$ direction.

$$s = \vec{v} \cdot \hat{e} + \vec{\beta} \cdot \hat{e} = \vec{v} \cdot \hat{e} - \beta \frac{x_0}{r_0}$$

max when $\vec{v}$ is in the same direction of $\hat{e}$

$$s = \left(v - \beta \frac{x_0}{r_0}\right)\hat{e}$$

$$t_1^* - t_0 = \frac{r_0}{v - \beta \frac{x_0}{r_0}} = \frac{r_0^2}{v r_0 - \beta x_0}$$

i) $v > \beta$

Since $\frac{x_0}{r_0} \le 1 \therefore x_0 \le r_0$, $v - \beta > 0$

$-\beta x_0 \ge -\beta r_0$
$v r_0 - \beta x_0 \ge v r_0 - \beta r_0 = r_0 (v - \beta)$
$v r_0 - \beta x_0 \ge r_0 (v - \beta) > 0 \quad \therefore \quad v r_0 - \beta x_0 > 0$

So denominator is always larger than zero. No restriction on x_0

ii) $v = \beta$

$v - \beta = 0, \quad x_0 \le r_0$
$v r_0 - \beta x_0 \ge r_0 (v - \beta) = 0$

Not reachable when denominator is zero. This happens iff $r_0 = x_0 \Rightarrow y_0 = 0$
$x_0 \ge 0$

iii) $v < \beta$

$v - \beta < 0, \quad x_0 \le r_0$
$v r_0 - \beta x_0 \ge r_0 (v - \beta) < 0 \quad \therefore \quad v r_0 - \beta x_0 > 0$

Origin reachable if $x_0 < \frac{v}{\beta} r_0$

1.1(g)

The problem can be written as a QP

$$\min_x \frac{1}{2}x^T Qx - b^T x \qquad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad Q = \begin{bmatrix} 1 & 0 \\ 0 & \gamma \end{bmatrix} \qquad b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since Q is positive definite, the optimal solution is

$$x^\star = Q^{-1}b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The constant-step plots use $\eta = 0.15$. The level sets satisfy

$$x_1^2 + \gamma x_2^2 = c$$

For $\gamma = 10$, these are narrow ellipses, so the curvature in the x_2 direction is much larger than in the x_1 direction. This causes zig-zagging for both exact line search and constant-step gradient descent.

For constant-step gradient descent,

$$x^{(k+1)} = (I - \eta Q)x^(k)$$

so

$$x_2^{(k+1)} = (1 - \eta\gamma)x_2^{(k)}$$

Zig-zagging begins when $1 - \eta\gamma < 0$. Convergence requires $|1 - \eta\gamma| < 1$, which gives $0 < \eta < 2/\gamma$. Then, for $\gamma = 10$, $\eta = 0.15$ zig-zags but still converges, while steps larger than 0.2 diverge. For $\gamma = 1$, the level sets are circular and exact line search reaches the optimum in one step.

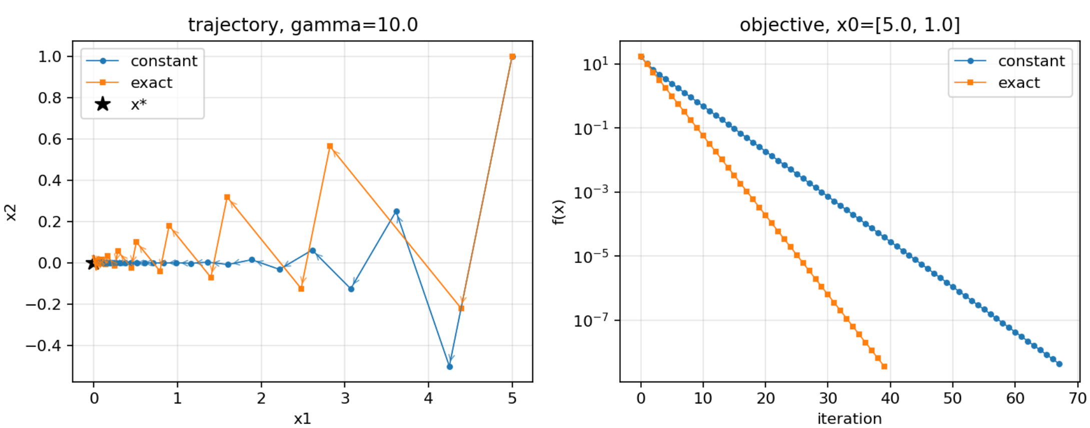


Figure 1: $\gamma = 10$, $x^{(0)} = (5, 1)$.

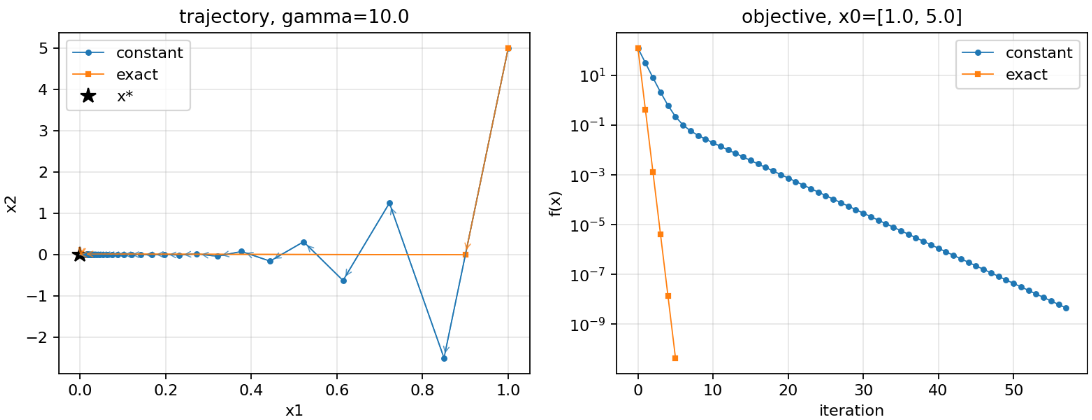


Figure 2: $\gamma = 10$, $x^{(0)} = (1, 5)$.

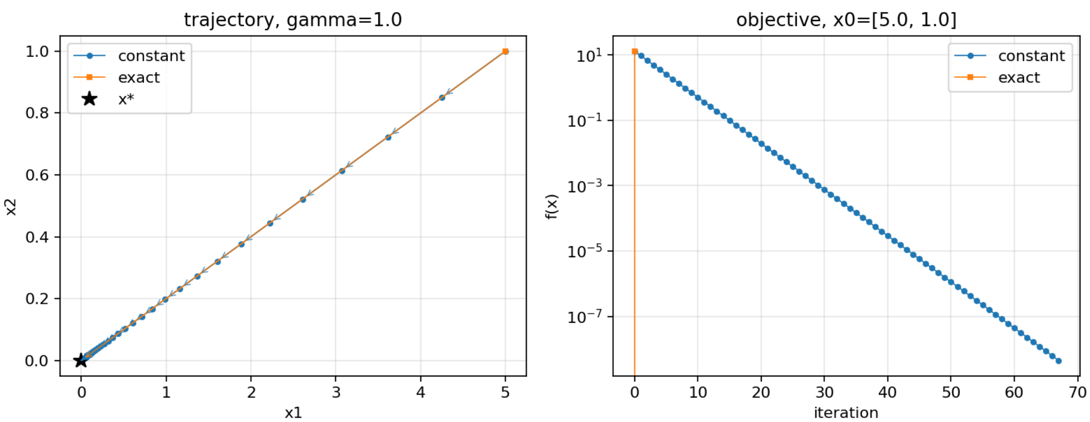


Figure 3: $\gamma = 1$, $x^{(0)} = (5, 1)$.

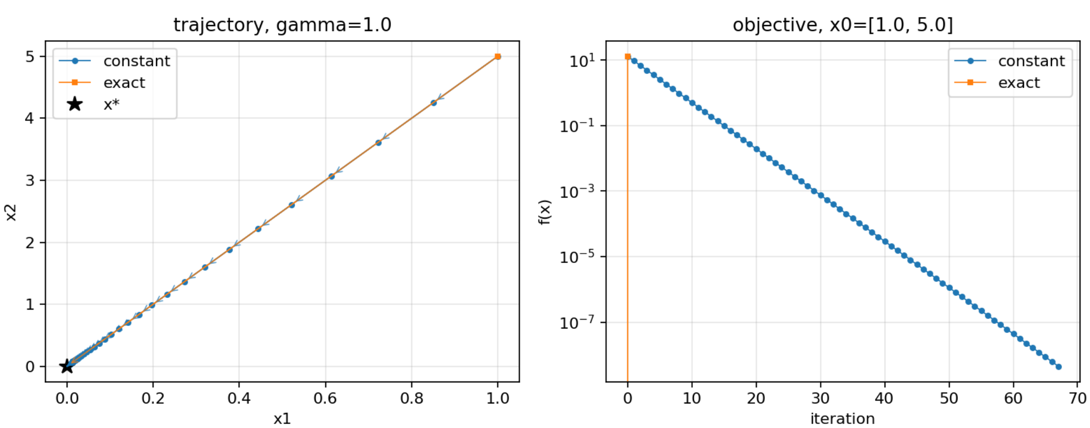


Figure 4: $\gamma = 1$, $x^{(0)} = (1, 5)$.

Code:

```
def f(x, q, b):
    return 0.5 * x.T @ q @ x - b.T @ x
```

```python

def grad_f(x, q, b):
    return q @ x - b

def gd_constant(x0, q, b, eta):
    x = x0.astype(float).copy()
    traj = [x.copy()]
    vals = [f(x, q, b)]
    tol = 1e-4

    while np.linalg.norm(grad_f(x, q, b)) > tol:
        g = grad_f(x, q, b)
        x = x - eta * g
        traj.append(x.copy())
        vals.append(f(x, q, b))

    return np.array(traj), np.array(vals)

def gd_exact_line_search(x0, q, b):
    x = x0.astype(float).copy()
    traj = [x.copy()]
    vals = [f(x, q, b)]
    tol = 1e-4

    while np.linalg.norm(grad_f(x, q, b)) > tol:
        d = -grad_f(x, q, b)
        eta = (d.T @ d) / (d.T @ q @ d)
        x = x + eta * d
        traj.append(x.copy())
        vals.append(f(x, q, b))

    return np.array(traj), np.array(vals)

def run(gamma, x0):
    q = np.diag([1.0, gamma])
    b = np.zeros(2)
    eta_const = 0.15
    traj_const, vals_const = gd_constant(x0, q, b, eta_const)
    traj_ls, vals_ls = gd_exact_line_search(x0, q, b)
```

1.2(b)

Using constant-step gradient descent with step size $\eta = 10^{-5}$.

The terminal output from the implementation is:

```
eta_const = 1.000000e-05
iterations = 279802
final objective = -27.052877
LQR cost J(u*) = 2.947123
```

Code:

```
T = 20
A = np.array([[1.0, 1.0], [0.0, 1.0]])
B = np.array([[0.0], [1.0]])
Q = np.eye(2)
R = np.eye(1)
Q_T = 10.0 * np.eye(2)
x0 = np.array([1.0, 0.0])

n = A.shape[0]
m = B.shape[1]
u0 = np.zeros(m * T)

nu = np.zeros((n * T, n))
V = np.zeros((n * T, m * T))

a_powers = [np.eye(n)]
for _ in range(T):
    a_powers.append(a_powers[-1] @ A)

for t in range(T):
    nu[t*n:(t+1)*n, :] = a_powers[t + 1]
    for k in range(t + 1):
        V[t*n:(t+1)*n, k*m:(k+1)*m] = a_powers[t - k] @ B

H = np.zeros((n * T, n * T))
for t in range(T - 1):
    H[t*n:(t+1)*n, t*n:(t+1)*n] = Q
H[(T - 1)*n:T*n, (T - 1)*n:T*n] = Q_T

N = np.zeros((m * T, m * T))
for t in range(T):
    N[t*m:(t+1)*m, t*m:(t+1)*m] = R

Q_tilde = 2.0 * (V.T @ H @ V + N)
b_tilde = -2.0 * V.T @ H @ nu @ x0

eta_const = 1e-5
traj_const, vals_const = gd_constant(u0, Q_tilde, b_tilde, eta_const)

u_star = traj_const[-1]
x_stack = nu @ x0 + V @ u_star
J_lqr = x0.T @ Q @ x0
for t in range(T - 1):
    x_t = x_stack[t*n:(t+1)*n]
    u_t = u_star[t*m:(t+1)*m]
    J_lqr += x_t.T @ Q @ x_t + u_t.T @ R @ u_t
x_T = x_stack[(T - 1)*n:T*n]
J_lqr += x_T.T @ Q_T @ x_T
```